# STAT 527 Advanced Regression Analysis
# Fall 2024

## Problems with the Predictors

**Jingbo Liu**

University of Illinois, Urbana-Champaign

# Problems with the Predictors

- Errors in the predictors (see chap 7 in Faraway "linear models with R"; we skip this)

- Collinearity (this talk)

# Collinearity

- If the $X$ variables in the sample are uncorrelated, the problem of choosing which one to include in the model is greatly simplified.
- In that case, $\beta_j$, the regression coefficient for $X_j$, does not depend on whether or not some other variable $X_k$ is included in the model.
- We often encounter data sets in which many of the predictors are highly correlated. Then the values and sampling variances of regression coefficients can be highly dependent on which particular predictors are in the model.
- This is called the problem of **collinearity** or **multicollinearity**.

# Collinearity

Two predictors $X_1$ and $X_2$ are **exactly collinear** if

$$c_1 X_1 + c_2 X_2 = c_0,$$

for some constants $c_1, c_2$, and $c_0$.

For example, suppose we wish to predict how many students $(Y)$ will skip class, as a function of temperature. To be careful, we use both $X_1$=temperature in Fahrenheit, and $X_2$=temperature in Celsius.

Then $X_1 - 1.8X_2 = 32$.

# Collinearity

This regression of $Y$ on $X_1$ and $X_2$ would involve a predictor matrix $\boldsymbol{X}$ with rows of the form

$$\begin{pmatrix} 1 & X_1 & X_2 \end{pmatrix}$$

and we would have

$$\begin{pmatrix} 1 & X_1 & X_2 \end{pmatrix} \begin{pmatrix} -32 \\ 1 \\ -1.8 \end{pmatrix} = -32 + X_1 - 1.8X_2 = 0$$

for *every* row of $\boldsymbol{X}$.

Thus, $\boldsymbol{X}$ would not have full (column) rank, $\boldsymbol{X}^T\boldsymbol{X}$ would not have an inverse, and we could not compute

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}.$$

# Approximate Collinearity

The definition of exact collinearity extends to $p$ predictors: for some constants $c_0, c_1, \ldots, c_p$,

$$\sum_{j=1}^{p} c_j X_j \;=\; c_0$$

Unless we have made some mistake as a data analyst, we generally do not have exact collinearity. However, we often must deal with **approximate collinearity**.

Approximate collinearity holds when at least one $X_k$ can be well approximated by the others,

$$X_k \;\approx\; \left(c_0 - \sum_{j \neq k} c_j X_j\right) \Big/ c_k.$$

# Collinearity Diagnostic

**Recall hw 1**

A simple diagnostic for this is to obtain the regression of $X_j$ on the remaining predictors, and record the coefficient of determination $R_j^2$ (that is, the $R^2$ for this regression).

If there is at least one $R_j^2$ close to 1, we would diagnose approximate collinearity.

# Why is approximate collinearity a problem?

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Let $r_{12}$ be the sample correlation between $X_1$ and $X_2$.

Then

$$\text{var}(\hat{\beta}_j) = \sigma^2 \left( \frac{1}{1 - r_{12}^2} \right) \left( \frac{1}{\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2} \right)$$

for $j = 1, 2$.

So the sampling variance becomes arbitrarily large as $|r_{12}| \to 1$.

**this follows from the general p case on the next slide (why?)**

# Variance Inflation

In the case of $p > 2$ the formula becomes

$$\text{var}(\hat{\beta}_j) = \sigma^2 \left( \frac{1}{1 - R_j^2} \right) \left( \frac{1}{\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2} \right)$$

The quantity $1/(1 - R_j^2)$ is called the $j$th **variance inflation factor**, denoted $VIF_j$.

A variable that is strongly affected by approximate collinearity will have a large $VIF$.

Rule of Thumb: A $VIF > 10$ indicates problems.

# Proof of the Variance Formula

- Recall $\text{var}(\hat{\beta}_j) = \sigma^2[(X^\top X)^{-1}]_{jj}$ It suffices to show that

$$\frac{1}{1-R_j^2} \cdot \frac{1}{\|X_j - \bar{X}_j\|^2} = [(X^\top X)^{-1}]_{jj}$$

- Since $\frac{\|X_j - \bar{X}_j\|^2}{\|X_j - \hat{X}_j\|^2} = \frac{1}{1-R_j^2}$, it boils down to

$$\frac{1}{\|X_j - \hat{X}_j\|^2} = [(X^\top X)^{-1}]_{jj} \tag{1}$$

- However, this was shown in hw1 problem 3 bonus question.

# Savings Data

Consider again the savings data ...

```
> library(faraway)  # provides the vif function

> mod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data=savings)

> vif(mod)
   pop15    pop75      dpi     ddpi
5.937661 6.629105 2.884369 1.074309
```

So no indication of severe collinearity problems.

# Eigenvalues of Gram Matrix

A global measure of collinearity is given by examining the eigenvalues of $\boldsymbol{X}^T\boldsymbol{X}$.

A popular measure is the *condition number* of $\boldsymbol{X}^T\boldsymbol{X}$, denoted by $\kappa$:

$$\kappa = (\text{largest eigenvalue/smallest eigenvalue})^{1/2}$$

Rules for declaring collinearity based on $\kappa$ have been proposed, such as declaring collinearity if $\kappa \geq 30$, but this is rather arbitrary and lacks theoretical justification.

# What to do about collinearity?

- If we mostly care about prediction, one option is to drop highly correlated predictors
- Variable selection may be used (will be discussed)
- If interpretation is important and you must keep all predictors, least squares not recommended. Some other estimation method, e.g., ridge regression (will be discussed)

# Thank you!

We gratefully acknowledge the previous course instructor Naveen Narisetty and various online sources for the images and slides presented.